

World's first commercial quantum computer beats conventional computer

A controversial machine bought in by Google and NASA, and named the world's first commercial quantum computer, has beaten a conventional computer in a series of tests, according to researchers at Google.

They also said that they had proof of it using quantum physics to work through a type of maths that is crucial to artificial intelligence (AI) and is much faster than a conventional computer.

Governments and leading computing companies such as Microsoft, IBM and Google are trying to develop what are called quantum computers because using the weirdness of quantum mechanics to represent data could unlock immense data-crunching powers. Computing giants believe quantum computers could make their AI software much more powerful and unlock scientific leaps in areas like materials science.

NASA hopes quantum computers could help schedule rocket launches and simulate future missions and spacecraft. The computer is installed at NASA's Ames Research Centre and operates on data using a superconducting chip called a quantum annealer, which is hard-coded with an algorithm suited to what are called optimisation problems. These are common in machine-learning and AI software.



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Fire-fighting robot that can shoot water, foam

A remote-controlled fire-fighting robot, known as Turbine Aided Fire-fighting machine or TAF 20, was recently unveiled in Sydney. It is capable of shooting a stream of water at 90m (295-feet), as well as spraying foam or mist at 60m (197-feet). The robot has a high-powered fan that can clear smoke from a room.

The robot will aid fire

and rescue NSW fire-fighters while managing hazardous fires and other emergencies where they cannot safely approach the flames, for example, when there is a danger of explosion and bushfires.

The robot will be based at Alexandria in inner Sydney and can be quickly deployed by road or air when and to where it is needed.



Fire-fighting robot